



MINING DEALER BEST PRACTICE

Pipe Rack Interlock Switch on MD6290, MD6240 and MD6420 Models

Maintenance and
Repair

Application

Component Life
Management

Component
Renewal (CRC)

MARC
Management

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1.0 Introduction

The machines models MD6420, MD6290 and MD6240 is equipped with a pipe rack interlock system. This system consists of a position switch – part number 517-7109 – installed on the mast to identify the position of pipe rack. If the pipe rack moves off its stop inside the mast the proximity switch will activate and stop the pulldown. This prevents the rotary head from hitting the pipe rack, preventing operational accidents.

To activate the pulldown command, it is necessary to swing the pipe rack against stop (open position). In that position the pulldown can resume.

But, when adding drill pipe however, you need to use pulldown and rotation with the pipe loader in the mast. To do this, place the switch in the operation's cab in the by-pass position (towards the unlock symbol). This will allow the rotation and pulldown functions to work when the pipe loader is out. The pictures 1 and 2 show the switches cited above.



Picture 1: Position switch on the mast



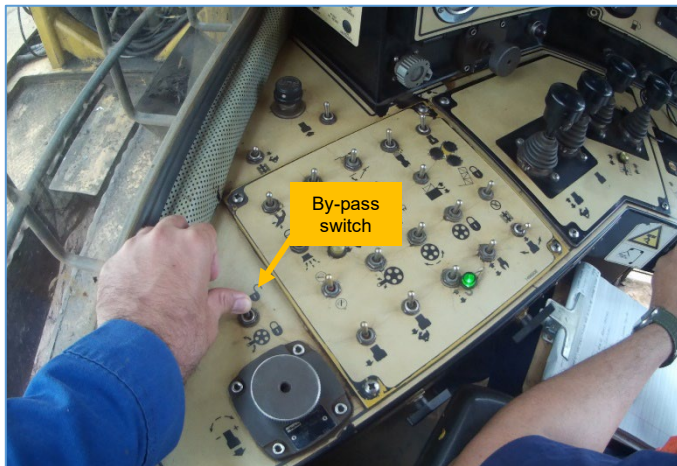
Picture 2: By-pass switch in the operator's cab

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2.0 Best Practice Description

The by-pass switch in the operator's cab (part number 472-4601) is a single two position switch and the position it is not returnable. At this condition, the operator can forget the switch in by-pass position and actuate the pulldown command with the pipe rack inside of the mast, causing an accident between the pipe rack and rotary drive.

To avoid this accident, the type of the by-pass switch was changed to a spring returnable switch part number 246-8912, as shown in the pictures 3 and 4.



Picture 3: Placing the switch in by-pass position



Picture 4: Switch returned to the original position

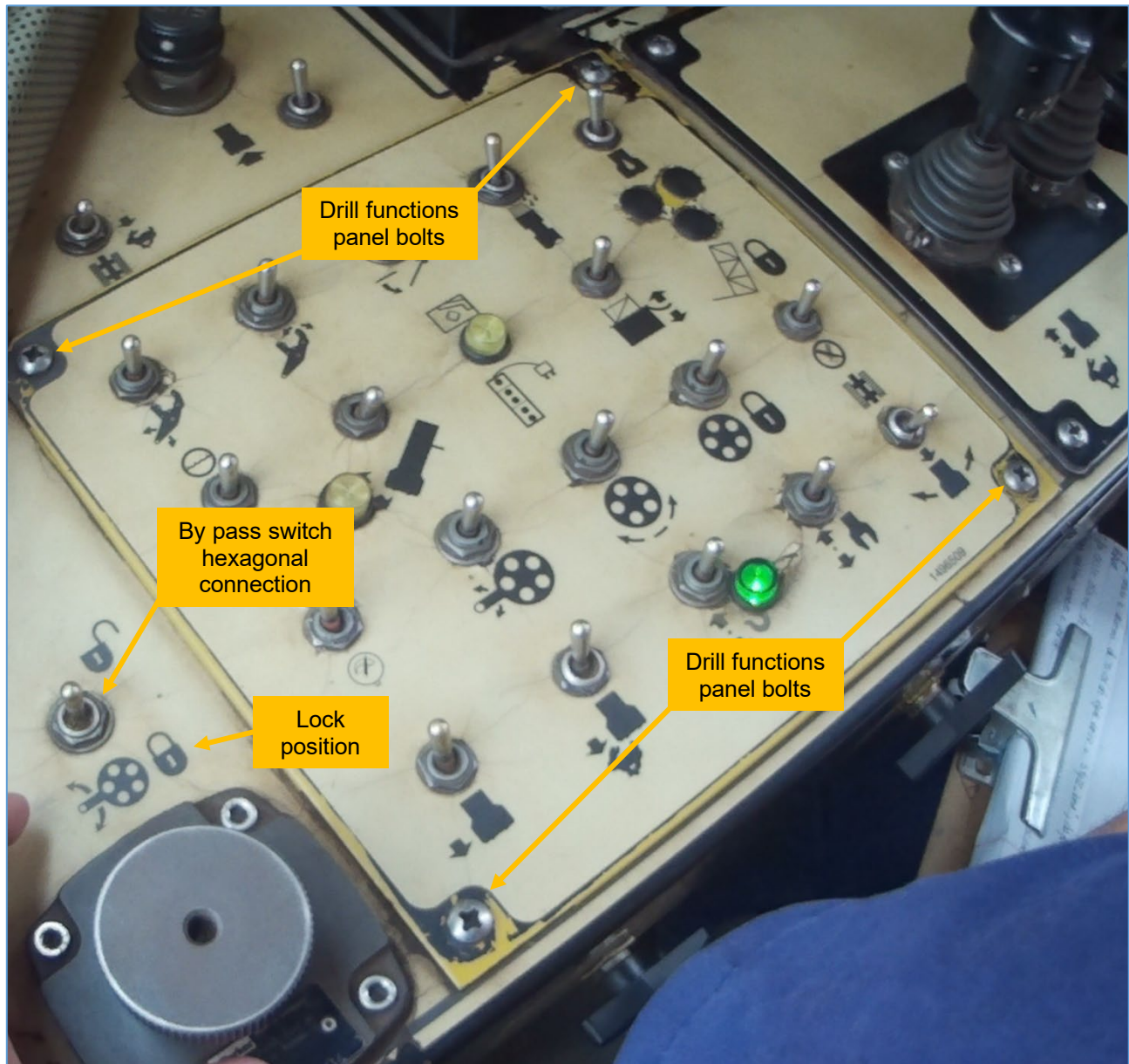
3.0 Implementation Steps

Note: Parts list in the "5.0 Resources Required"

1. Stop and shutdown the machine. Lock the safety device preventing switch, apply a warning sign.
2. Using a screwdriver, remove the installation bolts of the drill functions panel.
3. Access the electrical terminals of the by-pass switch and disconnect them.
4. Using a 1/2" open-end wrench at the hexagonal connection of switch, remove the original by-pass switch.
5. Using the same open-end wrench, install the new by-pass switch. Note: the normal position of this switch is in the lock symbol.
6. Connect the electrical terminals.
7. Using a screwdriver, install the drill functions panel.
8. Open the safety device preventing switch and turn on the machine.
9. Test the new by-pass switch installed in the pipe rack interlock system.

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The picture 5 shows details of this installation.



Picture 5: Details of change the by-pass switch

4.0 Benefits

1. Safety: Eliminate accidents by operator forget the switch at by-pass position.
2. Maintenance: Eliminate failures due to collisions rotary head and pipe rack.
3. Installation: It's a simple installation, with low labor.
4. Costs: Very low cost.

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5.0 Resources Required

The tables 1 and 2 below show the items and tolls used for this improvement.

Table 1: PARTS LIST			
ITEM	PART NUMBER	DESCRIPTION	QUANTITY
1	246-8912	Switch assembly	01 ea

Table 2: TOOLS		
TOOL	DESCRIPTION	QUANTITY
1	Screwdriver	01 ea
2	Open-end wrench	01 ea

Total parts cost US\$: 58.62

Labor: 0.5 hour – 1 technician

6.0 Supporting Attachments / References

BI012798-00 – Parts catalogue MD6290

7.0 Related Best Practices

N/A

8.0 Acknowledgements

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